

Marvin Ivan C. Mangubat

mic.mangubat17@gmail.com | Laguna/Manila | [LinkedIn](#) | [GitHub](#)

EDUCATION

De La Salle University - Manila
BSMS (Honors) Computer Science

Aug 2028
Malate, Metro Manila

- **CGPA:** 3.6 (Magna Cum Laude Standing); Consistent Dean's Lister
- **Coursework:** Machine Learning, Intelligent Systems, Full-Stack Development, Human-Computer Interaction, Theory of Computation

AI RESEARCH & PRODUCT DEVELOPMENT

AI-Driven Metagenomic Pipeline for Antimicrobial Resistance | *Master's Thesis*

- Architecting a hybrid Generative AI pipeline (ProtGPT2, ESM-2) to engineer novel phage lysins, directly addressing the high-friction clinical challenge of multi-drug resistant *Klebsiella pneumoniae*.
- Designing computational validation workflows integrating AlphaFold2 and GROMACS to extract interpretable latent design principles for non-natural sequences.

Machine Learning Lead: Phage-Enzyme Classifier | *Machine Learning Research Project*

- Developed an end-to-end ML pipeline using the ESM-2 Protein Language Model in Python and Jupyter Notebook, engineering high-dimensional embeddings from 2,500+ sequences (PhaLP 2.0) to classify phage lytic proteins for therapeutic applications.

PROFESSIONAL EXPERIENCE

Shopee | *Operations (Projects and Analytics) Intern*

Sep 2025 – Dec 2025

- Cut transcript review time by >95% and extracted actionable business insights by architecting a scalable Python-LLM analytics pipeline.
- Secured operational finances and mitigated reporting risk with 100% data accuracy by engineering a scalable Python data reconciliation pipeline that matched ~6,000 claims across 15+ datasets.
- Automated end-to-end data standardization and reporting to enable daily customer satisfaction dashboards for real-time operational visibility.

Unbound Manila Foundation Inc. | *Scrum Master (Client Project)*

May 2025 – Sep 2025

- Directed a cross-functional development team of 7 engineers using the MERN stack to architect and deliver a centralized case management system serving ~100 users.
- Managed sprint life cycles and translated partner requirements into actionable technical sprints, facilitating stakeholder feedback and integrating iterative systems design improvements to guarantee on-time delivery of features that resolved core operational bottlenecks.

LEADERSHIP & STRATEGY

DLSU Legislative Assembly | *Batch Legislator*

2024 – 2025

- Standardized academic integrity across all university faculties by authoring and implementing the institution's comprehensive ethical Generative AI policy.

DLSU SPRINT | *Associate Vice President for Externals*

2024 – Present

- Scaled organizational reach and professional network access for the student body by spearheading strategic partnerships with the Lasallian Scholars' Society and the DLSU Job Expo.
- Bridged the gap between academic theory and industry practice for organizational members by architecting 'SPRINT STATION,' a dedicated opportunity platform and speaking series.

Computer Studies Government | *Director for Relations*

2025 – Present

- Generated high-leverage external professional opportunities for the student body by forging cross-functional alliances with student organizations, industry leaders, and university offices.

TECHNICAL SKILLS

- **Technologies:** Python, Java, JavaScript, C, SQL, MongoDB, React, Node.js, Express, Git, Jupyter Notebook, Software Development, Testing, Systems Design
- **Core Competencies:** Artificial Intelligence, Machine Learning, Bioinformatics, Cross-Functional Leadership